

Received 15 February 2026, accepted 4 March 2026, date of publication 16 March 2026, date of current version 6 April 2026.

Digital Object Identifier 10.1109/ACCESS.2026.3674148

RESEARCH ARTICLE

IMPACT-Agent: Intelligent Multimodal Prompt-Driven Agent for Collaborative Teaming

TAEWAN CHO, HYUNWOO YOU, AND ANDREW JAEYONG CHOI^{ID}

School of Computing, Gachon University, Seongnam-si, Gyeonggi-do, South Korea

Corresponding author: Andrew Jaeyong Choi (andrewjchoi@gachon.ac.kr)

This work was supported by Gachon University Research Fund under Grant GCU-202500670001.

ABSTRACT The future of disaster response, search and rescue (SAR) technology is rapidly evolving with the integration of Artificial Intelligence (AI) and intelligent unmanned platforms. Manned-Unmanned Teaming (MUM-T) is revolutionizing on-site disaster structure and emergency rescue strategies by synchronizing manned and unmanned vehicles to support operations. This paper investigates the potential applications of Multimodal Large Language Models (MLLMs) in intelligent MUM-T systems. We present a system capable of using aerial imagery to identify and locate survivors and hazards/risks, and to generate safe path plans for unmanned ground vehicles (UGVs). The main contributions are: 1) the implementation of a Few-Shot In-Context Learning (ICL) method for MLLMs, which eliminates the need for fine-tuning or re-training the AI model, and 2) the application of the Chain-of-Thought (CoT) prompting technique. The ICL method reduces the time required for data gathering and model training, while the CoT reasoning technique improves the path planning generated by MLLMs. Our experiments, using aerial imagery, demonstrate that the proposed system achieves up to 86% mean Average Precision (mAP) in survivor and hazard identification through 5-shot ICL. Furthermore, applying the CoT technique doubled the UGV's path generation success rate from 20% to 40%. The proposed MLLM-based system effectively performed open-vocabulary object detection without additional training. This research highlights the potential of MLLMs for diverse and complex disaster relief missions in intelligent MUM-T systems, presenting an innovative approach to enhance capabilities and mission success while reducing human risk and operational costs. A demonstration video of our system is available at https://youtu.be/vReaSMDWu8o?si=RluQsqS4O_j5IVcS

INDEX TERMS Search and rescue (SAR), manned-unmanned teaming (MUM-T), observe, orient, decide, act (OODA) loop, multimodal large language models (MLLMs), few-shot in-context learning (ICL), Chain-of-Thought (CoT) prompting, human-machine collaboration.

I. INTRODUCTION

The future of disaster response, search and rescue (SAR) is rapidly expanding with the integration of Artificial Intelligence (AI) and intelligent unmanned platforms. Among these advancements, Manned-Unmanned Teaming (MUM-T) is revolutionizing disaster relief and SAR strategies by synchronizing manned and unmanned platforms to support operations, thereby reducing human risk and operational

costs [1]. MUM-T systems are designed to enable manned and unmanned platforms to cooperate effectively, performing complex and high-risk missions more efficiently and safely. A key goal in modern MUM-T for emergency scenarios is accelerating the decision-making cycle, often conceptualized by the Observe, Orient, Decide, Act (OODA) Loop framework. Recent research has explored integrating new AI paradigms, such as Large Language Model (LLM)-based operating systems, to enhance this cycle [2], [3].

Specifically, SAR operations are fundamentally challenging due to unstructured, hazardous, and time-critical

The associate editor coordinating the review of this manuscript and approving it for publication was Domenico Rosaci^{ID}.

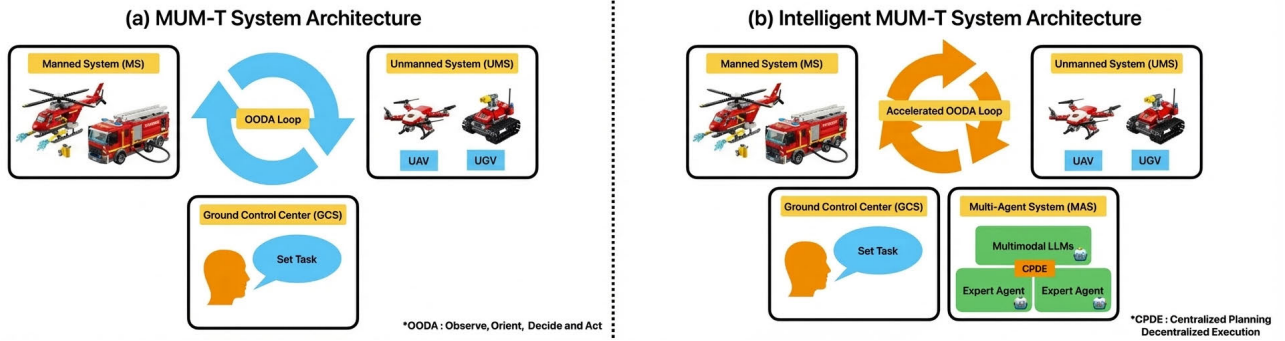


FIGURE 1. Comparison between traditional and the proposed intelligent MUM-T system architectures. (a) Traditional MUM-T system with a basic OODA loop. (b) The proposed intelligent MUM-T system featuring an accelerated OODA loop and a Multi-Agent System (MAS) with MLLMs and expert agents.

environments [4]. The integration of UAVs has become a key strategy to overcome these challenges, offering rapid situational awareness and access to inaccessible areas. A recent comprehensive survey highlights the state-of-the-art in UAV platforms, sensor integration, and autonomous algorithms tailored for SAR missions [5]. Our work builds upon this foundation by proposing an MLLM-based agent architecture to further enhance the cognitive and collaborative capabilities of these systems.

The challenge of enhancing human-machine teaming is a critical area of research, with recent studies demonstrating the potential of foundation models to improve human-robot collaboration (HRC) in complex tasks like manufacturing assembly [6]. Similarly, current MUM-T systems face significant challenges. These include data bottlenecks from processing large volumes of real-time data, leading to slower decision-making, and a heavy reliance on the cognitive abilities of human operators, which increases the risk of misjudgment in complex scenarios. In response, studies on multimodal learning [7], [8], [9], multi-agent reinforcement learning [10], and LLM-based reward design [11] have begun to address the limitations of traditional architectures and improve autonomous decision-making.

However, while these approaches show promise, they often rely on extensive fine-tuning with domain-specific datasets, which is impractical for rapidly changing disaster and emergency scenarios. A critical research gap exists in leveraging the zero-shot or few-shot capabilities of foundational models for MUM-T without the need for costly retraining. This paper directly addresses this gap by investigating the application of Multimodal Large Language Models (MLLMs) to intelligent MUM-T systems to enhance decision-making and operational efficiency in disaster relief and SAR missions. In this paper, we focus on MLLMs, a class of foundation models capable of processing various data modalities. This category includes Vision-Language Models (VLMs), which specifically handle text and images, the primary focus of our system. By processing and integrating diverse data types, MLLMs offer a more comprehensive understanding of dynamic environments compared to

traditional single-modal models. This capability is crucial for complex decision-making and can significantly accelerate the OODA loop in MUM-T systems.

Fig. 1 illustrates a conceptual comparison between a traditional MUM-T system and the proposed intelligent MUM-T architecture based on IMPACT-Agent. As shown in Fig. 1(a), conventional MUM-T systems typically follow a basic OODA loop, where perception, decision-making, and control are loosely coupled and heavily reliant on human operators. This structure often results in delayed responses and increased cognitive burden, particularly in complex and time-critical disaster scenarios.

In contrast, Fig. 1(b) presents the proposed architecture, which integrates Multimodal Large Language Models (MLLMs) within a Multi-Agent System (MAS) to accelerate the OODA loop. By enabling MLLMs to function as high-level reasoning and coordination agents, the proposed system facilitates tighter coupling between perception, decision-making, and action. Specialized expert agents handle task-specific execution, while the MLLM provides semantic understanding, reasoning, and explainable decision support. This architectural shift highlights how IMPACT-Agent enhances situational awareness, reduces operator workload, and enables more adaptive and collaborative human-machine teaming in disaster response environments.

We propose a system, IMPACT-Agent, that can identify and locate survivors and hazards and generate safe navigation paths for unmanned ground vehicles (UGVs) using only aerial imagery. The main contributions of this paper are twofold: (1) the implementation of the Few-Shot In-Context Learning (ICL) method for MLLMs, which eliminates the need for model fine-tuning or retraining, and (2) the application of the Chain-of-Thought (CoT) prompting technique. The remainder of this paper is organized as follows:

- 1) Section II reviews related works concerning MUM-T and MLLMs.
- 2) Section III describes the proposed IMPACT-Agent architecture and detailed algorithmic processes.

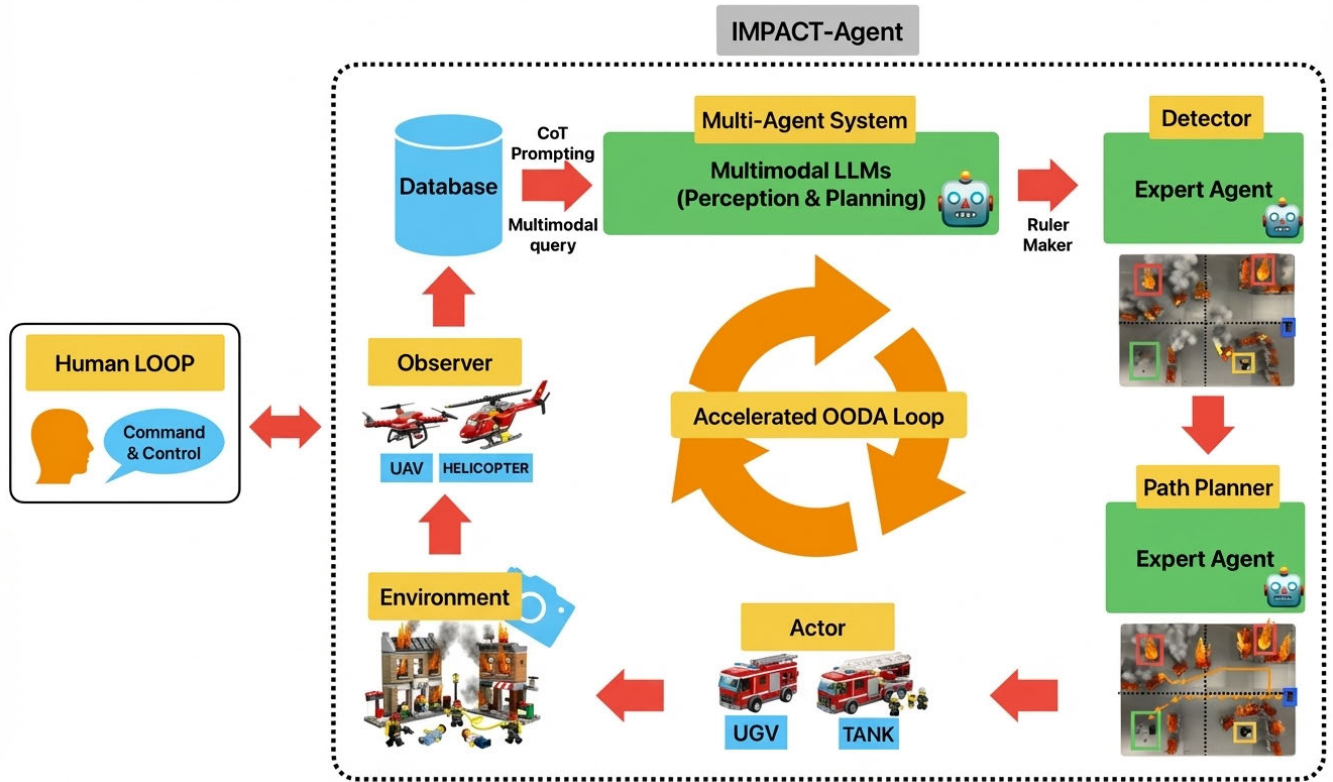


FIGURE 2. Overview of the IMPACT-Agent architecture for an accelerated OODA loop in MUM-T systems. The system integrates a human operator, a multi-agent system with MLLMs, and specialized agents (Detector, Path Planner) to process environmental data and control assets (UAV, UGV, TANK).

- 3) Section IV presents the experimental setup, RQs, and analysis of results.
- 4) Section V discusses the implications and limitations of the findings.
- 5) Section VI concludes the paper.

II. RELATED WORKS

A. RESEARCH ON MUM-T AND OODA LOOP ACCELERATION

MUM-T is a concept aimed at achieving operational effectiveness by integrating heterogeneous platforms. As defined by the U.S. Army Aviation Center of Excellence (USA ACE), MUM-T systems maximize collaboration to enhance situation awareness, survivability, and mission completion rate by enabling flexible mission execution. Key challenges include real-time processing of large-scale data, reducing operator cognitive load, and ensuring interoperability. The OODA loop is a fundamental framework, and current research focuses on sensor fusion, decision support systems, and reinforcement learning to improve its efficiency. However, limitations in handling unstructured data and the risk of operator misjudgment persist. To overcome these, studies on data-centric strategies, explainable AI, and autonomous platform collaboration are being spotlighted. For instance, research on remote sensing automation [12] and multimodal prompting [13], [14] offer promising avenues for improving

situational awareness. Unlike these studies that focus on sensor fusion or reinforcement learning within traditional architectures, our work explores how MLLMs can serve as a central reasoning engine to accelerate the entire decision-making loop.

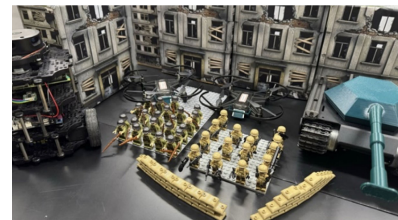


FIGURE 3. Components for the scale disaster zone experimental environment. The setup includes toy figures representing survivors and hazard markers, along with UGV and destroyed tank models.

B. MLLMs FOR PERCEPTION AND CONTROL IN AUTONOMOUS SYSTEMS

Recent advancements in deep learning, particularly the Transformer architecture [15], have established a paradigm of pre-training on large datasets. To mitigate the high costs and challenges of task-specific fine-tuning, methods such as Retrieval-Augmented Generation (RAG) [16] and ICL have emerged. Foundational models like CLIP [17]

demonstrate strong zero-shot performance, while the CoT prompting technique [18] enhances step-by-step reasoning. These capabilities have been successfully applied to complex perception tasks like Open-Vocabulary Detection (OVD) using models such as GLIP [19] and Grounding DINO [20]. Concurrently, MLLMs are being leveraged for autonomous control, with systems like GPT-Driver [21] and MobilityVLA [22] interpreting multimodal commands for navigation. While these studies highlight the broad potential of MLLMs [23], our work is distinct in its application of these fine-tuning-free methods to the unique challenges of a disaster response and SAR MUM-T context, focusing on rapid and reliable deployment in dynamic scenarios.

III. SYSTEM ARCHITECTURE

To overcome the challenges of traditional MUM-T systems, we propose a Multi-Agent System (MAS) architecture, named IMPACT-Agent, based on MLLMs. This architecture enhances situational awareness and decision-making performance in disaster and SAR operations without additional model training by integrating specialized AI agents with MLLMs. The proposed architecture accelerates the OODA loop compared to traditional systems, as illustrated in Fig. 2. The architecture is composed of four key modules:

1. **Sensor and Data Acquisition Module (UAV-based Observation):** UAVs capture real-time disaster site conditions and preprocess the raw aerial imagery before feeding it to MLLMs and specialized AI agents. This module handles large-scale data collection and enhances situational awareness in the Observe and Orient phases of the OODA loop.

2. **Multimodal LLM-Based Situational Awareness Module:** MLLMs combine textual queries and aerial imagery to identify key objects (survivors, hazards, obstacles) and describe the emergency situation in natural language. Leveraging Few-Shot ICL and Ruler Marker techniques improves the model's adaptability and accuracy without additional training.

3. **Specialized AI Agent Integration Module:** This module translates high-level semantic plans from the MLLM into executable commands. Specifically, the Path Planner Agent receives grid-based waypoints from the MLLM (e.g., "Move to C4") and converts them into precise vector coordinates for the UGV's navigation stack using A* or similar pathfinding algorithms, ensuring collision-free trajectory generation.

4. **Decision-Making and Collaboration Support Module:** MLLMs apply the CoT method to explicitly reveal their reasoning processes. By providing understandable explanations, the human operator can trust, and if needed, intervene or adjust decisions, enabling flexible and explainable decision-making.

The overall workflow of the IMPACT-Agent, which interacts with various specialized AI agents to perform its tasks, is depicted in Fig. 2. Algorithm 1 details the operational logic of the proposed system.

Algorithm 1 IMPACT-Agent Decision-Making Process

```

1: Input: Aerial Image  $I_{raw}$ , Mission Goal  $G$ 
2: Output: Navigation Path  $P$ , Rationale  $R$ 
3:  $I_{grid} \leftarrow \text{ApplyRulerMarker}(I_{raw})$  {Overlay  $10 \times 10$  grid}

4:  $Prompt \leftarrow \text{Combine}(G, I_{grid}, \text{Few-Shot Examples})$ 
5: if CoT Enabled then
6:    $[R, P_{text}] \leftarrow \text{MLLM\_Generate\_with\_CoT}(Prompt)$ 
7: else
8:    $P_{text} \leftarrow \text{MLLM\_Generate}(Prompt)$ 
9: end if
10:  $P_{coord} \leftarrow \text{ParseCoordinates}(P_{text})$ 
11:  $P_{final} \leftarrow \text{PathPlannerAgent}(P_{coord})$  {Collision Avoidance}
12: return  $P_{final}, R$ 

```

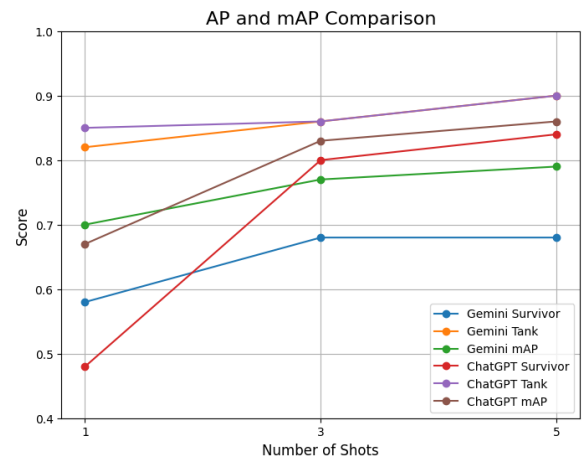


FIGURE 4. Performance comparison using Few-Shot ICL with two MLLMs. The graph shows the Average Precision (AP) for survivor and tank detection, and the mean Average Precision (mAP) for both Gemini and GPT-4o models across 1, 3, and 5 shots. The upward trend for both models confirms that performance scales effectively with the number of in-context examples provided.

IV. EXPERIMENTS

A. EXPERIMENTAL SETUP

To evaluate the performance of the proposed MLLM-based IMPACT-Agent system, we constructed a lab-scale disaster response environment (Fig. 3). The experiments were designed to answer two Research Questions (RQs):

- **RQ1:** Can MLLMs accurately identify survivor locations, obstacles, and hazards based on the aerial imagery captured by a quadcopter?
- **RQ2:** Are the paths generated by the MLLMs reliable for the UGV to effectively avoid collisions? Does utilizing the CoT approach improve path generation performance?

To mimic real emergency conditions, a scenario was designed in which survivors are isolated within a dangerous zone and need to be rescued. The disaster zone was represented using toy blocks for survivors and hazard markers. A total of

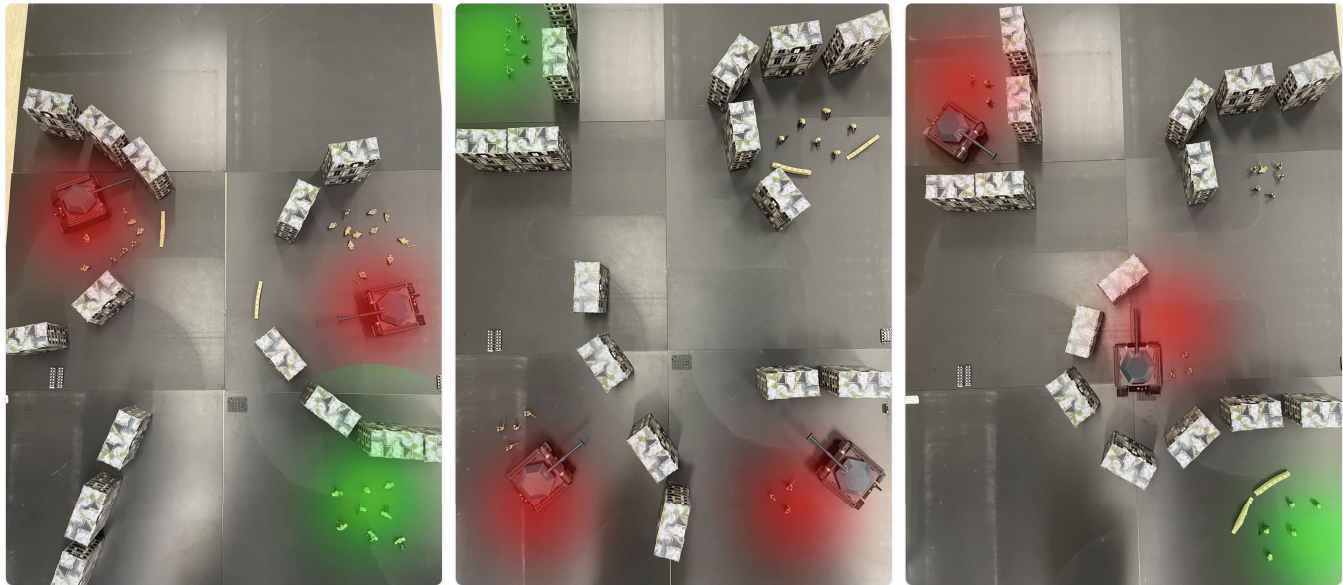


FIGURE 5. Results of RQ1 Experiment to identify and locate both survivors and hazards in three different layouts. The images show the model's detection results with bounding boxes for survivors (green) and hazards (red).

14 survivor toy figures, 24 hazard markers, one rescue UGV, and one destroyed vehicle model were placed.

B. EXPERIMENT AND ANALYSIS FOR RQ1

In this experiment, we aimed to determine whether the proposed MLLMs could identify survivors and hazards and their locations using ICL without additional training. Two backbone models, Gemini 1.5 Pro and GPT-4o, were used. Prompts were structured to include the scenario description, role assignment, image description, and task explanation. Using a few-shot prompting technique with 1, 3, and 5 shots, we verified that performance improved with more examples, as shown in Fig. 4. The results with 5-shot learning, presented in Fig. 5 and Table 1, show that both models demonstrated high performance in situational awareness tasks when the Intersection over Union (IoU) threshold was set to 0.3. Furthermore, applying the Ruler Marker technique (Fig. 6) improved mAP accuracy by approximately 9%. This experiment demonstrated that general-purpose models can be effectively applied to specialized tasks, emphasizing the versatility and adaptability of the proposed MLLM-based approach.

TABLE 1. Situational awareness experimental results by model (5-Shot).

Detection Category	Evaluation Metric	Gemini 1.5 Pro	GPT-4o
Number of Survivors	MAE	1.62	1.24
Number of Tanks	MAE	0.04	0.08
Number of Hazards	MAE	2.36	2.70
Location of Survivors	AP	0.68	0.84
Destroyed Tanks Location	AP	0.90	0.90
Overall Location	mAP	0.79	0.86

C. EXPERIMENT AND ANALYSIS FOR RQ2

The RQ2 experiment evaluated whether the MLLM could independently control the UGV using only aerial imagery.

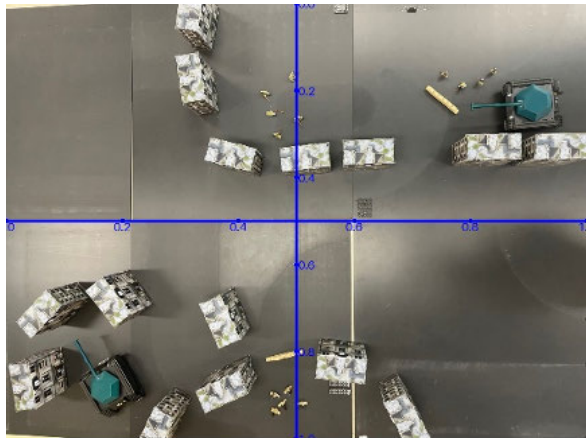


FIGURE 6. Applying the ruler marker technique on the aerial imagery to increase object localization. The grid overlay helps the MLLM to better understand and describe the spatial positions of objects.

TABLE 2. UGV path planning success rate without Chain-of-Thought.

Trials	Gemini 1.5 Pro	GPT-4o
1	X	X
2	X	X
3	X	O
4	X	X
5	O	X
6	X	X
7	X	X
8	X	O
9	O	X
10	X	X
Success Rate	20%	20%

Initially, without CoT, both models succeeded in only 2 out of 10 trials (20% success rate), as shown in Table 2. To overcome this, we implemented the CoT technique, which guides the

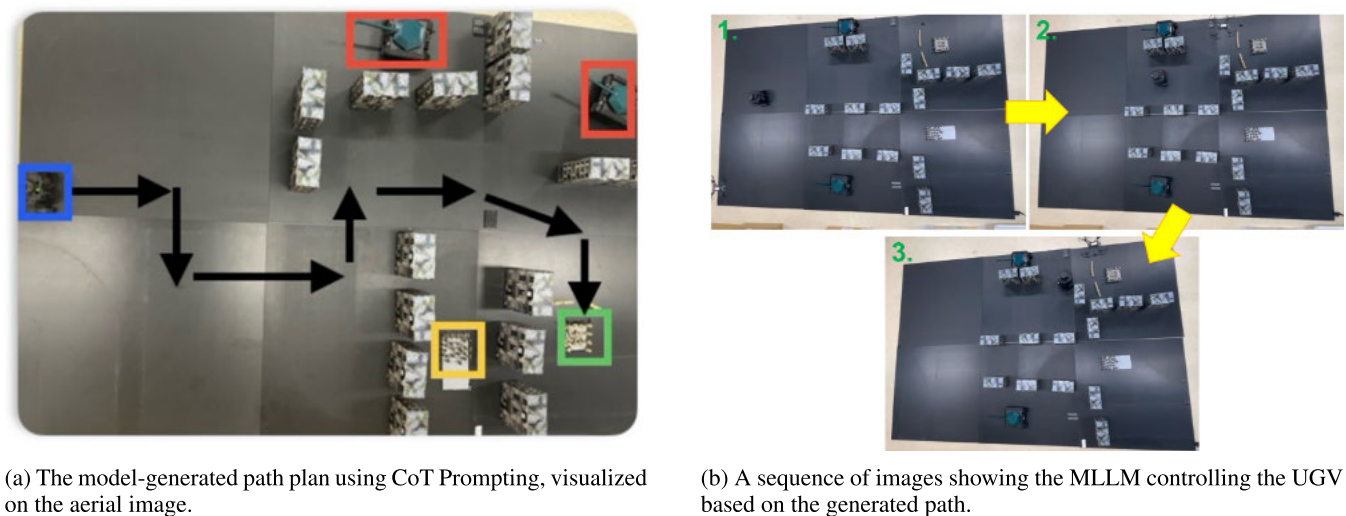


FIGURE 7. Result of RQ2 experiment to generate path planning for the UGV.

TABLE 3. Path planning success rate doubled with Chain-of-Thought.

Trials	Gemini 1.5 Pro	GPT-4o
1	X	X
2	X	O
3	O	X
4	O	X
5	X	O
6	X	X
7	X	X
8	O	O
9	O	O
10	X	X
Success Rate	40%	40%

Based on the drone reconnaissance image, the survivor is located in the 2nd zone, assuming the accessible zones are numbered from left to right. Zone 1 harbors hazards/risks, while zones 3 and 4 are occupied by damaged structures. Therefore, to avoid risks/obstacles and safely rescue the survivor, the UGV should follow this path:

FIGURE 8. The reasoning process generated by Gemini using CoT prompting. The model explicitly states its logic for choosing a specific path to avoid hazards.

MLLM to describe its reasoning before finalizing the path. As a result, the success rate for path generation doubled to 40% (Table 3). The CoT technique improved both the interpretability and accuracy of the generated paths. Fig. 7 shows the generated path and UGV navigation. An analysis of Gemini’s reasoning (Fig. 8) showed it accurately identified the target location in 7 out of 10 trials, demonstrating the

potential of CoT for robust reasoning even when the final navigation fails.

V. DISCUSSION

Our experimental results provide critical insights into both the potential and the current limitations of applying MLLMs to intelligent MUM-T systems. This discussion interprets these findings, contextualizes them within the broader field, and outlines their implications for future research. The success of the ICL method in our situational awareness task (RQ1) demonstrates a potential paradigm shift for deploying AI in rapidly evolving disaster and emergency environments. Traditional deep learning approaches require extensive, domain-specific datasets and time-consuming fine-tuning cycles. Our findings suggest that general-purpose MLLMs can achieve competent performance on specialized tasks like survivor and hazard identification with only a handful of examples, drastically reducing the time from data acquisition to operational deployment. The notable performance increase of approximately 9% in mAP from the Ruler Marker technique further underscores a key insight: augmenting visual data with simple, structured information can significantly enhance an MLLM’s spatial grounding capabilities, acting as a cognitive aid to bridge the gap between pixel space and semantic understanding.

The results from our path planning experiment (RQ2) offer a more nuanced perspective. While the application of CoT prompting doubled the success rate from 20% to 40%, highlighting its effectiveness in promoting structured, step-by-step reasoning, the overall success rate remains modest for safety-critical applications. A qualitative analysis of the failure cases reveals that they primarily stem from two sources: (1) perceptual errors, where the MLLM occasionally misidentified obstacles or safe passages despite

the CoT process, and (2) suboptimal spatial logic, where the generated path was locally coherent but globally inefficient or led to a dead end. This indicates that while MLLMs excel at high-level semantic reasoning (e.g., “avoid the major hazard”), they still face challenges in precise, low-level geometric and spatial reasoning required for robust navigation.

Our findings provide compelling evidence for the necessity of the hybrid, MAS architecture proposed in this paper (Fig. 1b). Rather than tasking an MLLM with end-to-end control, a more robust approach is to leverage it as a high-level commander or reasoning engine. In this role, the MLLM interprets the disaster scene context, sets strategic goals (e.g., “rescue the survivors at grid C4”), and provides a rationale for its decisions, thereby enhancing human-on-the-loop collaboration. The execution of precise tasks, such as object detection with high IoU thresholds or generating collision-free trajectories, should be delegated to specialized expert agents (e.g., a fine-tuned YOLO model or a classical A* path planner). This division of labor harnesses the semantic understanding of MLLMs while relying on the proven reliability of specialized algorithms for mission-critical functions. This collaboration structure exemplifies effective Human-Machine Collaboration, where the AI augments human cognitive capabilities while specialized agents ensure safety.

In particular, the proposed IMPACT-Agent framework exemplifies *Human-Machine Collaboration* by tightly integrating human oversight with machine-driven perception, reasoning, and execution. While Multimodal Large Language Models (MLLMs) and specialized expert agents autonomously perform situation understanding, task planning, and coordination, human operators remain actively involved in supervising high-level goals, validating critical decisions, and intervening when necessary. This collaborative interaction enables the system to leverage the complementary strengths of humans and machines, combining human contextual judgment and accountability with the scalability, speed, and consistency of intelligent agents.

Unlike fully autonomous systems, the proposed hybrid Multi-Agent System (MAS) is explicitly designed to support shared authority and responsibility between human operators and machine agents. As a result, IMPACT-Agent not only enhances operational efficiency and adaptability in complex disaster response scenarios but also provides a concrete architectural realization of Human-Machine Collaboration, justifying its inclusion as a key concept of this study.

This conclusion aligns with an emerging consensus in the field. Recent studies, such as Cooper et al. [24], have also re-evaluated the roles of large models like VLMs and LLMs, suggesting that while they are powerful reasoners, specialized models often remain superior for specific perception tasks. Therefore, the true power of MLLMs in systems like IMPACT-Agent lies not in replacing every component,

but in acting as an intelligent orchestrator that integrates various capabilities, communicates with human operators in natural language, and brings a new level of cognitive flexibility to autonomous systems.

A. ETHICAL CONSIDERATIONS

While the integration of MLLMs into MUM-T systems significantly enhances operational efficiency and decision-making speed, it necessitates a rigorous discussion on Responsible AI. The deployment of autonomous systems in disaster response, search and rescue (SAR) must strictly adhere to ethical guidelines to mitigate risks associated with algorithmic bias and hallucination [25]. As demonstrated in our findings, while MLLMs serve as high-level reasoners, they are not infallible. Therefore, ensuring Meaningful Human Control (MHC) is imperative. The human operator must remain the ultimate decision-maker, particularly in lethal operations, to satisfy International Humanitarian Law (IHL) and ensuring accountability [26]. Furthermore, the “black box” nature of deep learning models poses challenges to transparency. Our application of Chain-of-Thought (CoT) prompting not only improves performance but also enhances explainability, allowing operators to verify the AI’s logic before execution. This aligns with the principle that AI in high-stakes environments must be transparent and traceable to prevent “automation bias,” where operators overly trust AI recommendations without critical evaluation [27]. Consequently, the IMPACT-Agent architecture is designed not to replace human judgment but to augment it, ensuring that the pursuit of autonomous capabilities does not compromise ethical standards and human safety.

B. PRIVATE USE CASES AND SOCIAL VALUES

The system can be applied directly to the following civilian applications: (i) detection of survivors at flood and earthquake sites and securing routes for affected areas; (ii) remote reconnaissance and access warnings in hazardous chemical leak areas; and (iii) congestion monitoring and emergency response support at large events. IMPACT-Agent provides natural language-based explanations and justifications through MLLM, allowing operators to quickly understand and make judgments on the basis of recommendations.

VI. CONCLUSION

This paper presented the capability of an MLLM-based system to identify and locate survivors and hazards and generate safe UGV paths from aerial imagery. The main contributions—the implementation of the ICL method and the application of the CoT prompting technique—were validated through experiments. The proposed intelligent MUM-T system demonstrated its potential to accelerate the OODA loop and facilitate efficient collaboration between manned and unmanned platforms in disaster relief operations.

However, the experiments also exposed inherent accuracy constraints. This suggests that MLLMs are best utilized as effective communication interfaces and high-level reasoners within a broader MAS, where specialized agents handle task-specific functions to ensure precision and reliability. This hybrid approach, where MLLMs act as high-level reasoners rather than end-to-end controllers, aligns with the emerging consensus on the optimal role for large models in complex systems [24].

Future research will focus on applying the MAS in additional scenarios, integrating sensor fusion, reinforcement learning, and swarm intelligence methodologies. Additionally, advanced safety verification frameworks for autonomous UGVs [28], distributed adaptive cooperative control for heterogeneous UAV-UGV systems [29], and dynamic task allocation strategies using graph neural networks [30] can be leveraged to further enhance reliability, scalability, and mission success rates. Such an approach will pave the way for more robust, reliable, and adaptable operations in increasingly complex environments. This study provides a foundation for formulating AI utilization strategies that advance the sophistication of intelligent MUM-T systems.

REFERENCES

- [1] A. A. Bazin, "Boyd's O-O-D-A loop and the infantry company commander," *Infantry*, vol. 94, no. 1, pp. 17–19, Jan. 2005.
- [2] J. Ge, C. Chang, J. Zhang, L. Li, X. Na, Y. Lin, L. Li, and F.-Y. Wang, "LLM-based operating systems for automated vehicles: A new perspective," *IEEE Trans. Intell. Vehicles*, vol. 9, no. 4, pp. 4563–4567, Apr. 2024.
- [3] L. Li, Y. Li, X. Zhang, Y. He, J. Yang, B. Tian, Y. Ai, L. Li, A. Nüchter, and Z. Xuanyuan, "Embodied intelligence in mining: Leveraging multi-modal large language models for autonomous driving in mines," *IEEE Trans. Intell. Vehicles*, vol. 9, no. 5, pp. 4831–4834, May 2024.
- [4] R. R. Murphy, *Disaster Robotics*. Cambridge, MA, USA: MIT Press, 2014.
- [5] M. Lyu, Y. Zhao, C. Huang, and H. Huang, "Unmanned aerial vehicles for search and rescue: A survey," *Remote Sens.*, vol. 15, no. 13, p. 3266, Jun. 2023.
- [6] Y. Ji, Z. Zhang, D. Tang, Y. Zheng, C. Liu, Z. Zhao, and X. Li, "Foundation models assist in human-robot collaboration assembly," *Sci. Rep.*, vol. 14, no. 1, p. 24828, Oct. 2024.
- [7] Z. Lan, L. Liu, B. Fan, Y. Lv, Y. Ren, and Z. Cui, "Traj-LLM: A new exploration for empowering trajectory prediction with pre-trained large language models," *IEEE Trans. Intell. Vehicles*, pp. 1–14, 2024.
- [8] Y. Tian, X. Li, H. Zhang, C. Zhao, B. Li, X. Wang, X. Wang, and F.-Y. Wang, "VistaGPT: Generative parallel transformers for vehicles with intelligent systems for transport automation," *IEEE Trans. Intell. Vehicles*, vol. 8, no. 9, pp. 4198–4207, Sep. 2023.
- [9] M. Adil, H. Song, S. Mastorakis, H. Abulkasim, A. Farouk, and Z. Jin, "UAV-assisted IoT applications, cybersecurity threats, AI-enabled solutions, open challenges with future research directions," *IEEE Trans. Intell. Vehicles*, vol. 9, no. 4, pp. 4583–4605, Apr. 2024.
- [10] Y. Xue and W. Chen, "Multi-agent deep reinforcement learning for UAVs navigation in unknown complex environment," *IEEE Trans. Intell. Vehicles*, vol. 9, no. 1, pp. 2290–2303, Jan. 2024.
- [11] X. Han, Q. Yang, X. Chen, Z. Cai, X. Chu, and M. Zhu, "AutoReward: Closed-loop reward design with large language models for autonomous driving," *IEEE Trans. Intell. Vehicles*, pp. 1–13, 2024.
- [12] K. Wu, Y. Zhang, L. Ru, B. Dang, J. Lao, L. Yu, J. Luo, Z. Zhu, Y. Sun, J. Zhang, Q. Zhu, J. Wang, M. Yang, J. Chen, Y. Zhang, and Y. Li, "A semantic-enhanced multi-modal remote sensing foundation model for Earth observation," *Nature Mach. Intell.*, vol. 7, no. 8, pp. 1235–1249, Aug. 2025.
- [13] J. Wang, A. Ma, Z. Chen, Z. Zheng, Y. Wan, L. Zhang, and Y. Zhong, "EarthVQANet: Multi-task visual question answering for remote sensing image understanding," *ISPRS J. Photogramm. Remote Sens.*, vol. 212, pp. 422–439, Jun. 2024.
- [14] H. Zhu, B. Zhou, Z. Lu, X. Sui, Y. Zhao, Y. Zhang, and X. Yuan, "Coarse-to-fine multimodal prototype network for few-shot multimodal relation extraction," *Knowl.-Based Syst.*, vol. 330, Nov. 2025, Art. no. 114569.
- [15] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, Ł. Kaiser, and I. Polosukhin, "Attention is all you need," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 30, 2017, pp. 1–12.
- [16] P. Lewis, E. Perez, A. Piktus, F. Petroni, V. Karpukhin, N. Goyal, H. Küttler, M. Lewis, W.-T. Yih, T. Rocktäschel, S. Riedel, and D. Kiela, "Retrieval-augmented generation for knowledge-intensive NLP tasks," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 33, 2020, pp. 9459–9474.
- [17] A. Radford, J. W. Kim, C. Hallacy, A. Ramesh, G. Goh, S. Agarwal, G. Sastry, A. Askell, P. Mishkin, J. Clark, G. Krueger, and I. Sutskever, "Learning transferable visual models from natural language supervision," in *Proc. 38th Int. Conf. Mach. Learn. (ICML)*, vol. 139, 2021, pp. 8748–8763.
- [18] M. Bosma, E. Chi, B. Ichter, Q. V. Le, D. Schuurmans, X. Wang, J. Wei, F. Xia, and D. Zhou, "Chain-of-thought prompting elicits reasoning in large language models," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 35, 2022, pp. 24824–24837.
- [19] L. H. Li, P. Zhang, H. Zhang, J. Yang, C. Li, Y. Zhong, L. Wang, L. Yuan, L. Zhang, J.-N. Hwang, K.-W. Chang, and J. Gao, "Grounded language-image pre-training," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 10965–10975.
- [20] S. Liu, Z. Zeng, T. Ren, F. Li, H. Zhang, J. Yang, Q. Jiang, C. Li, J. Yang, H. Su, J. Zhu, and L. Zhang, "Grounding DINO: Marrying DINO with grounded pre-training for open-set object detection," in *Proc. Comput. Vis. ECCV*, in Lecture Notes in Computer Science, vol. 15105. Springer, 2024, pp. 38–55.
- [21] J. Mao, Y. Qian, J. Ye, H. Zhao, and Y. Wang, "GPT-driver: Learning to drive with GPT," 2023, *arXiv:2310.01415*.
- [22] Z. Xu, H.-T. L. Chiang, Z. Fu, M. G. Jacob, T. Zhang, T.-W. E. Lee, W. Yu, C. Schenck, D. Rendleman, D. Shah, F. Xia, J. Hsu, J. Hoeck, P. Florence, S. Kirmani, S. Singh, V. Sindhwani, C. Parada, C. Finn, P. Xu, S. Levine, and J. Tan, "Mobility VLA: Multimodal instruction navigation with long-context VLMs and topological graphs," presented at the *Proc. 8th Conf. Robot Learn. (CoRL)*, vol. 270, 2025, pp. 3866–3887.
- [23] Y. Jiang, A. Gupta, Z. Zhang, G. Wang, Y. Dou, Y. Chen, L. Fei-Fei, A. Anandkumar, Y. Zhu, and L. Fan, "VIMA: Robot manipulation with multimodal prompts," in *Proc. 40th Int. Conf. Mach. Learn. (ICML)*, vol. 202, 2023, pp. 14975–15022.
- [24] A. Cooper, K. Kato, C.-H. Shih, H. Yamane, K. Vinken, K. Takemoto, T. Sunagawa, H.-W. Yeh, J. Yamanaka, I. Mason, and X. Boix, "Rethinking VLMs and LLMs for image classification," *Sci. Rep.*, vol. 15, no. 1, p. 19692, Jun. 2025.
- [25] V. Dignum, "Responsible artificial intelligence: Designing AI for human values," *ITU J., ICT Discoveries*, vol. 1, no. 1, pp. 1–8, 2018.
- [26] F. S. de Sio and G. Mecacci, "Four responsibilities for meaningful human control over artificial intelligence," *Nature Mach. Intell.*, vol. 3, no. 12, pp. 1057–1063, 2021.
- [27] A. Dass and M. Jain, "Ethics, transparency, and explainability in AI-based decision support systems," *IEEE Potentials*, vol. 41, no. 2, pp. 24–29, 2022.
- [28] D. Meltz and H. Guterman, "Functional safety verification for autonomous UGVs—Methodology presentation and implementation on a full-scale system," *IEEE Trans. Intell. Vehicles*, vol. 4, no. 3, pp. 472–485, Sep. 2019.
- [29] H. Liang, S. Yang, T. Li, and H. Zhang, "Distributed adaptive cooperative control for human-in-the-loop heterogeneous UAV-UGV systems with prescribed performance," *IEEE Trans. Intell. Vehicles*, vol. 9, no. 11, pp. 1–13, Nov. 2024.

- [30] Z. Ma, J. Xiong, H. Gong, and X. Wang, "Adaptive depth graph neural network-based dynamic task allocation for UAV-UGVs under complex environments," *IEEE Trans. Intell. Vehicles*, vol. 10, no. 5, pp. 3573–3586, May 2025.



HYUNWOO YOU is currently pursuing the master's degree in artificial intelligence with Gachon University, Seongnam. As a master's student, his work focuses on the intersection of computer vision and deep learning. In addition, he is interested in unmanned systems, such as drones, and is conducting research to incorporate AI into them.



TAEWAN CHO received the B.S. degree from the School of Computing, Gachon University, Seongnam, South Korea, in 2024, where he is currently pursuing the M.S. degree. His primary research focuses on developing novel architectures to unlock and interpret latent geometric knowledge within large-scale foundation models. His broader interests include the efficient adaptation of vision-language models and spatial perception for embodied AI.



ANDREW JAEYONG CHOI received the B.S. degree in electrical and computer engineering from Purdue University, West Lafayette, IN, USA, in 2014, the M.S. degree in electrical and computer engineering from North Carolina State University, Raleigh, NC, USA, in 2016, and the Ph.D. degree in aerospace engineering from Korea Advanced Institute of Science and Technology (KAIST), Daejeon, South Korea, in 2022. He is currently an Assistant Professor with the School of Computing, Gachon University. His current research interests include physical AI, artificial general intelligence (AGI) robotics, LLM agents, and robot vision.

• • •